

OFF-ROAD DRIVING AND SOIL COMPACTION. Name: Julian Zille

Name 3 factors that determine susceptibility of soils to crusting and sub soil compaction? (3)

- particle size, distribution, shape/smoothness (eg high fine-sand = extremely susceptible)
- organic matter content in soil (low = less stability, less buffer against compaction = suscep)
- parent material of soil. (intrusive igneous ^{sandstone} harder → finer ^{child} soil → more susc.)
(extrusive igneous opposite eg basalt)

Name 2 consequences of soil crusting (2)

1. Bare soil develops with no recovery / vegetative growth (no water/gas infiltration)
2. Increased soil erosion

How does soil crusting increase soil erosion (2)

- compacted soil →
- Lower infiltration rate →
- more surface runoff.
- Also, compact soil → less organic matter to ~~but~~ slow down runoff.

What are the consequences of sub surface soil compaction (2)

Roots are blocked from penetrating compacted layer ∴ can't reach water in subsoil.

What percentage of compaction is made from a driving over loose soil for the first pass and what percentage of sub soil compaction occurs? (2)

- (i) 90%.
- (ii) 60%.

Give 4 factors that will help ecological sustainability (4)

- eliminate bare soil (prevent overgrazing)
- control + restrict vehicle traffic
- smart (re)design of road networks ⇒ soil sensitivity map
- use appropriate (lower) tyre pressures.

Which three types of areas should be avoided at all costs with off-roading? (3)

Ramscar pans, Vlei areas, barren areas with no grass cover

When must hubs be locked and unlocked? (2)

Hubs must be locked ~~when~~ before going into H4/L4
unlocked before going to H2 For long trips on hard roads